



Tonal Perception in Changde Mandarin

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Abstract

This paper explores perceptual cues of the four tones in Changde Mandarin. Results show that T1 is perceived as a high-level tone, T2 is a low-rising tone in both production and perception, T3 is not phonologically level but falling, and the perception of T4 is influenced by various factors. And results also show that tonal perception might not be categorical, since neither the T2-T3 continuum nor the T2-T4 continuum fulfills the standards for typical categorical perception.

Index Terms: tone, perceptual cues, categorical perception, Changde Mandarin

1. Introduction

Many languages in the world are tonal [1]. Chinese is an exemplar that uses pitch or fundamental frequency (F0) to distinguish lexical meanings [2] [3]. The four lexical tones in Standard Chinese are transcribed as /55 35 214 51/ in Chao's tone letters [4]. Chinese tones have attracted abundance of research interest for decades. And there are two issues essential to tonal perception. The first one concerns perceptual cues of tone. In general, pitch, intensity, duration, and phonation all contribute to tonal perception [5]. And pitch could be further differentiated into relative height or register and the dynamic movement or contour. There is a long debate on the difference between T2 and T3 in Standard Chinese. Both tones have a phonetically concave shape and are easily misperceived possibly due to their similar pitch onset, contour shape, and duration [6] [7] [8] [9]. Some research proposed that T2 and T3 differ in concavity, with T2 having an earlier turning point than T3 [10] [11], while others treated T2 as rising and T3 as a phonological low tone [12] [13]. However, there are still others argue against the low tone account, because they found that native speakers favor a low concave pitch contour as an appropriate representation of T3 [11] [14] [15].

The second issue concerns the debate whether tones are perceived categorically. Typically, in categorical perception, (1) the peak in discrimination accuracy should align with the category boundary where rapid changes in identification probability occur, and (2) the discrimination accuracy should be predictable from the identification results [16] [17] [18] [19]. It is well documented that consonants are perceived categorically [20] [21] [22], whereas vowels are perceived in a continuous manner [23] [24] [25]. But there is no consensus regarding tone perception [26] [27] [28] [29] [30] [31].

This paper pursues these two theoretical issues by investigating Changde Mandarin, a regional variety of the south-western Mandarin dialects [32]. Changde is situated in the northwest of Hunan Province. Similar to Standard Chinese, Changde Mandarin also has four tones, but with different

phonetic tonal values in Chao's tone letters: /45 23 21 24/. Figure 1 shows pitch contours of the four tones and their 2-sigma ellipses in the pitch-slope plot. A typologically marked phenomenon is that Changde Mandarin has no level tone: T1 is high rising and T3 is low falling. It is thus essential to investigate whether the dynamics of F0 plays a role in the perception of T1 and T3, or they mainly distinguish by high versus low pitch. Another marked tonal phenomenon is that T2 and T4 have similar pitch onset and contour shape. Identification and discrimination experiments are designed to explore perceptual cues of the four tones in general and to examine the particular differences between tonal pairs with similar or different contours. The findings would shed light on the controversy as to whether tonal perception is categorical or not, and contribute to a better understanding on tonal contrasts in a tonal language with a contour tone system.

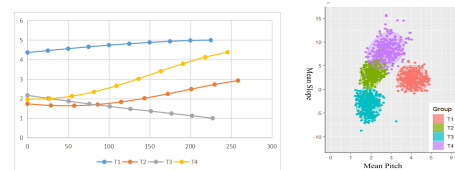


Figure 1: Mean pitch contours (left) and 2-sigma ellipses (right) for the tones in Changde Mandarin (data based on 12 speakers [33]).

2. Methodology

2.1. Subjects

31 native speakers (10 males and 21 females) were recruited. None of them reported any speech or hearing disorders. All subjects completed the experiments except that one male did not participate in Experiment 2.

2.2. Stimuli

2.2.1. Experiment 1

Target syllables /mi/ were provided by a male speaker with four tones: /mi⁴⁵/ “squint”, /mi²³/ “puzzle”, /mi²¹/ “rice”, /mi²⁴/ “honey”. Each syllable was embedded in a carrier sentence [mi, tau²³ mi ke²¹ ni²¹ thij⁴⁵] “mi, read mi for you to listen.” Four repetitions in a randomized order were recorded. The rhyme was labeled as the tone-bearing unit and annotated in Praat 6.0.29 [34]. Ten equidistant sampling points of F0 across the entire duration were extracted and measured. A typical sample for each tone was selected from the recordings as the model of the synthetic [mi] syllable. Table 1 shows the reference pitch values for each tone. Based on the reference pitch range, pitch

curves were generated in a five-scale register: low (L) at 110 Hz, half-low (hL) at 135 Hz, middle (M) at 160 Hz, half-high (hH) at 185 Hz, and high (H) at 210 Hz.

Table 1: Reference pitch values in Herz at 10 equidistant sampling points of four tones.

	1	2	3	4	5	6	7	8	9	10
T1	201	205	208	210	212	216	219	223	225	226
T2	115	116	118	118	119	122	124	129	135	140
T3	119	116	114	114	113	113	111	109	107	100
T4	128	127	127	132	139	148	159	171	183	192

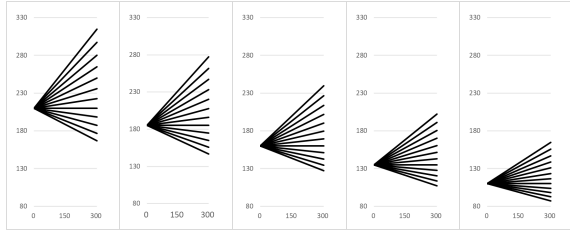


Figure 2: Synthesized pitch curves from falling to rising at five registers (from left to right: H, hH, M, hL, L) in Exp.1

To encapsulate the diverse shapes of tonal contours, our design manipulates the offset of each pitch curve by one semitone increment, enabling transitions from slightly descending to flat, then to low-rising, and ultimately to high-rising contours. Given that T4 exhibits the greatest pitch increase (approximately 7 semitones) and T3 the greatest decrease (around 4 semitones), we incorporated seven rising steps (S7 to S1) and four falling steps (S-1 to S-4) relative to each base pitch curve. As shown in Figure 2, we totally have a set of 60 stimuli (5 registers $\times$ 12 steps), with a normalized time of 300 milliseconds.

2.2.2. Experiment 2

The second experiment is designed to explore the differences between the two tonal pairs, namely T2-T3 and T2-T4. As shown in Figure 3, two 10-step pitch continua with equivalent step length in semitone distance are constructed through PSOLA re-synthesis in Praat: one is ranging from T2 to T3 (left), the other is ranging from T2 to T4 (right).

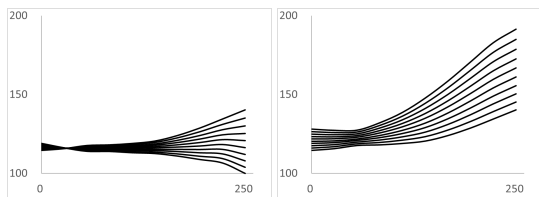


Figure 3: Synthesized pitch curves in two sets of continua (left: T2-T3; right: T2-T4) in Exp.2.

2.3. Procedure

2.3.1. Experiment 1

An identification task was conducted. Four Chinese characters, [mi] syllables with the four tones, were displayed on the screen. Upon hearing a stimulus [mi^x] (^x represents one of the four tones), the participant was required to identify it as one of the

four choices by pressing the corresponding key on the keyboard. A total of 240 stimuli (60 stimuli repeated four times) were presented in ten blocks with a randomized order. Participants' responses and reaction times were automatically recorded.

2.3.2. Experiment 2

Participants are required to complete an identification task first, and then a discrimination task. The procedure of the identification task is similar to Experiment 1. For each continuum, 40 stimuli (10 steps $\times$ 4 repetitions) are played in a random order for subjects to identify in 2 blocks.

Either ABX or AX paradigm is used in discrimination task. Of the 30 participants, 15 participants each completed both the AX and ABX tasks. For AX paradigm, stimuli were presented in pairs with a 500-ms interstimulus interval (ISI) to maximize differences in the performance of between- versus within-category discrimination ([36]). The whole 416 pairs (26 pairs $\times$ 4 repetitions) were presented in a random order. In each set, 16 pairs consist of two different stimuli separated by two steps in either forward 1-3, 2-4, 3-5...7-9, 8-10 or reverse order 3-1, 4-2, 5-3, etc; 10 pairs consist of two identical stimuli, like 1-1, 2-2, 3-3, etc. After hearing each pair, participants need to judge whether the two stimuli were the same or different, and to respond by pressing the key "g" (same) or "h" (different).

For ABX paradigm, three stimuli A, B, and X are played successively in each trial, where A and B are two stimuli one step apart on the continuum, such as S1-S3, S2-S4, S3-S5... The third stimulus, X, is either identical to A (e.g., S1-S3-S1) or to B (e.g., S1-S3-S3). After listening to the three stimuli, participants need to respond accordingly by pressing "g" for A or "h" for B as soon as possible. To counteract any sequence effects, all possible permutations of ABX are considered: ABA (e.g., S1-S3-S1), ABB (e.g., S1-S3-S3), BAA (e.g., S3-S1-S1), BAB (e.g., S3-S1-S3). Each sequence is repeated four times, resulting in a total of 128 trials (4 sequences $\times$ 8 pairs $\times$ 4 repetitions). The stimuli are presented in a random order.

3. Results

3.1. Experiment 1

3.1.1. Mean identification rate

Figure 4 shows the average response percentages of the 12-step tonal continua from falling to rising at 5 register groups.

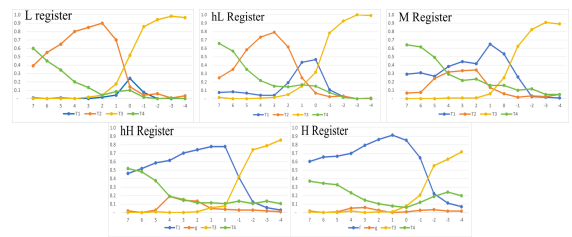


Figure 4: Mean identification rate at 12 steps in 5 register groups.

At the L register, the rising contours S1 to S7 are predominantly identified as T2 and T4. As the rising slope decreases, the recognition rate of T4 decreases and the rate of T2 steadily rises, peaking at 0.9 at S2. When the contour is level S0, the recognition of T2 drops sharply to the bottom, and there

is a 0.5 chance of it being identified as T3. And the falling contours S-1 to S-4 are predominantly recognized as T3, with a recognition rate ranging from 0.9 to 1.0. And the hL register has a similar perceptual pattern to the L register. In summary, low pitch contours are identified as T3 when the contour is falling, are identified as T2 when the contour is slightly rising, and are identified as T2 or T4 the contour is sharp rising. That is, it is difficult to identify T4 in the low pitch registers.

The M register also demonstrates a three-way perceptual pattern. When the contour is falling, it is again identified as T3. When the contour is sharp rising, T4 has the highest identification rate, but the competitive tone is T1 rather than T2. When the contour is slightly rising, the identification rates of T1, T2 and T4 are all below the chance level.

The two high registers hH and H exhibit a similar pattern, which is different from the low and mid registers. When the contour is rising or level, it is predominantly identified as T1, except that T4 is a competitor when the contour is sharp rising. When the contour is falling, it is likely identified as T3, except that when the falling is least at -1 step the identification of T1 could challenge or even prevail. And there is a systematic difference between the hH and H registers. That is, a higher pitch of H favors the identification of T1, and a relatively lower pitch of hH favors the identification of T3.

3.1.2. Statistical analysis

A generalized linear model was employed for each tonal category to fit the identification rate, setting Slope (pitch contour) and Register (pitch height) as independent variables. Results indicate that both pitch register and slope have significant influences on the recognition rate for all the four tones (T1-Register: wald chi-square=1367.5, $p<0.00$; T1-Slope: wald chi-square=865.24; $p<0.00$; T2-Register: wald chi-square=1091.61, $p<0.00$; T2-Slope: wald chi-square=1049.13; $p<0.00$; T3-Register: wald chi-square=280.80, $p<0.00$; T3-Slope: wald chi-square=5739.36; $p<0.00$; T4-Register: wald chi-square=24.39, $p<0.00$; T4-Slope: wald chi-square=528.66; $p<0.00$). And there exists an interaction between the two factors (T1: wald chi-square=361.30, $p<0.00$; T2: wald chi-square=886.63, $p<0.00$; T3: wald chi-square=346.65, $p<0.00$; T4: wald chi-square=130.93, $p<0.00$).

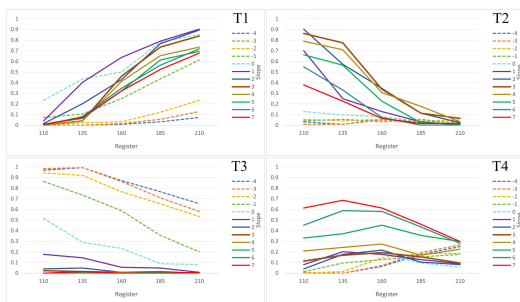


Figure 5: Estimated identification rates of the four tones by Glm models.

Figure 5 shows the predicted identification rate for each tone based on generalized linear models. First, T1 is sensitive to both pitch register and slope. When the pitch register is below 160 Hz, T1 has a low rate of identification. Additionally, when the pitch contour is falling (S-2 to S-4), T1 has a low rate of identification regardless of the pitch register. However, when the pitch contour is rising, level, or slightly falling, the

identification rate of T1 shows a strong positive correlation with the pitch register: higher pitch register is associated with a higher rate of identification.

Second, T2 has an identification pattern mirrored to T1. T2 has a low rate of identification when the pitch register is high or when the pitch contour is falling or level. And when the pitch contour is rising, the identification rate of T2 shows a negative correlation with the pitch register in general: higher pitch register is associated with a lower rate of identification. But it should be noted that the slope also plays a role: the identification of T2 prefers a slight rising contour, rather than a sharp rise.

Third, slope prevails register in the identification of T3. Only a falling contour has a high identification rate of T3. When the pitch contour is rising or even level, the identification rate is low regardless of pitch register. When the pitch contour is falling, the identification rate shows a negative correlation with the pitch register: higher pitch register is associated with a lower rate of identification. But the lowest identification rate is above the chance level except for the least falling contour (S-1).

Finally, T4 has the worst rate of identification, with a highest rate below 0.7. Only the pitch contour is sharp rising (S6 and S7) and the register is lower than 170 Hz, would T4 be the favored choice.

3.2. Experiment 2

As stated in Formula 1, a logistic regression generalized linear model was employed to determine the location of the category boundary [31]. This model employed the identification score (P1) and the number of Steps (S) as variables, where S, despite being sampled at discrete steps within a continuum, can be considered a continuous variable. Formulas 2-4 explained the detailed procedures (see also [21] [35] [36] for reference). Additionally, to investigate whether the AX task is easier for participants than the ABX task, we conduct a post-hoc comparison of the discrimination accuracy across different paradigms within each continuum.

$$\ln\left(\frac{P_1}{1-P_1}\right) = b_1 S + b_0 \quad (1)$$

$$P_{AX} = P(S) * P(S' | S) + P(D) * P(D' | D) \quad (2)$$

$$P_{ABX} = \frac{1}{4} (P(ABA) + P(ABB) + P(BAA) + P(BAB)) \quad (3)$$

$$P' = \frac{1 + (P_A + P_B)^2}{2} \quad (4)$$

3.2.1. T2-T3 continuum

Figure 6 depicted the measured (solid line) and predicted scores (dashed line) of identification task, and discrimination score predicted by Glm model (solid line) and identification result (calculated via Formula 6) of the T2-T3 continuum.

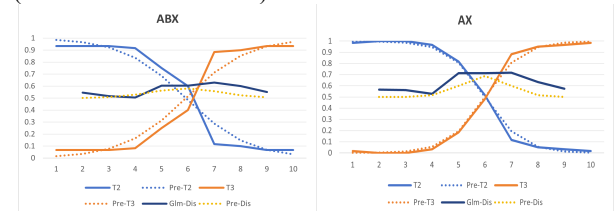


Figure 6: Identification and discrimination curves of the T2-T3 continuum (left: ABX; Right: AX).

First, both the measured and predicted identification curves for T2 and T3 exhibit distinct plateaus with high identification

rates ranging from 0.9 to 1.0. These plateaus indicate the presence of two clear categorical regions on the left and right sides of the continua. Notably, the predicted curves derived from logistic regression align well with the measured curves, particularly for the AX group. Second, although there is slight difference, the estimated discrimination accuracy curve generated by the Glm model generally surpasses the predicted discrimination curve in both groups. This suggests that participants' performance exceeded what the identification result could predict. And statistical results suggest that neither the ABX nor AX group displays the classical categorical perception pattern due to their poor performance on the discrimination task.

3.2.2. T2-T4 continuum

As depicted in Figure 7, the identification results for the T2-T4 continuum do not show a typical category pattern. Only the identification curve on the left side of the continuum in the AX group has an ideal identification rate for T4, spanning from 0.9 to 1.0. By contrast, the identification results for T2 are relatively poor, particularly in the AX group, with a lower rate of approximately 0.8 at the right side of the continuum.

Also, the discrimination curves for the T2-T4 continuum are around or below 0.6 and exhibit no discernible peak, regardless of the group or calculation method employed. Furthermore, the two discrimination curves overlap heavily with each other in the ABX group, whereas the estimated curve is marginally higher than the predicted curve in the AX group. Both indicate a low discrimination accuracy for the T2-T4 continuum.

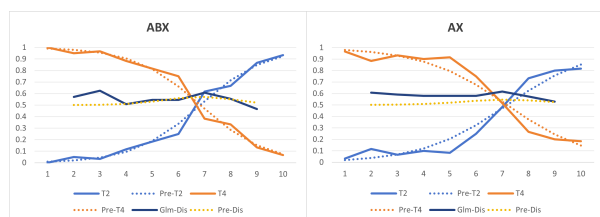


Figure 7: Identification and discrimination curves of the T2-T4 continuum (left: ABX; Right: AX).

4. Discussion and Conclusions

What is essential in tonal phonology is to explain how tones distinguish from each other. Among the four tones /45 23 21 24/ in Changde Mandarin, T1 is high, T2 and T3 are low, but T4 is ambiguous. How these tones are differentiated from each other in perception? Is tonal perception categorical? This paper employed two experiments to address these two issues in tonal perception in Changde Mandarin.

Experiment 1 aims to investigate perceptual cues of the four tones in Changde Mandarin. Results show that both pitch height/register and pitch contour/slope are critical for the identification of the tones in general on the one hand. On the other hand, register and slope play different roles in the perception of individual tones.

First, register is more important than slope in the perception of T1. As long as the register is sufficiently high, T1 is tolerant of slope. For instance, the sharpest rising contour S7 has an identification rate above 0.6, as shown in Figure 4. However, the slope is statistically significant, too. T1 prefers a slightly

rising or level contour, and the least rising contour (S1) and the level pitch curve (S0) have the highest rate of identification in the H and hH groups respectively in Figure 4. To conclude, T1 could phonologically be characterized as [+High] [+Level], with a phonetic realization as high level or high (slightly) rising.

Second, T3 is a low falling tone. But it could not be treated as a low tone phonologically, because slope is more important than register in perception. As evidenced in Figure 5, all falling contours have identification rates higher than chance level except the least falling slope (S-1), whereas all the rising contours have identification rates lower than 0.2. And even the level contours under the condition of low pitch register also have identification rates lower than chance level in general. Therefore, T3 should phonologically be characterized as [+Falling] [+Low].

Third, the two rising tones, T2 and T4, could not be differentiated perceptually by a different rising slope, as suggested by their production as well as their transcriptions [23] versus [24]. Rather, high rising pitch contours are likely to be identified as T2, no matter whether the rising slope is sharp or not. In short, T2 could phonologically be characterized as [+Low] [+rising]. But as for T4, the experiments did not yield satisfactory results. The results suggest that T4 prefers a low pitch register below 170Hz and a sharp rising contour (S6 and S7). But it should be admitted that further research is required to explore how T4 is distinguished from T2.

Experiment 2 employed both the ABX and AX paradigms to examine whether the homomorphic tonal continuum T2-T4 and the heterotonic tonal continuum T2-T3 are perceived categorically or not. Results show that the T2-T3 continuum exhibited an ideal identification pattern in the AX group: T2 and T3 achieved nearly 100% recognition rates within their respective perceived advantage intervals, and their identification scores were highly sensitive at the categorical boundary, showing sharp rises or falls and quickly transitioning to another categorical region. Although the identification performance in the ABX group was not as good as that of the AX group, it still presented two distinguishable categorical regions on both sides of the continuum. However, neither the ABX group nor the AX group demonstrated a corresponding discrimination peak at the category boundary, failing to meet the strictly defined criteria for categorical perception. Regarding the T2-T4 continuum, the results did not align with the criteria for categorical perception, regardless of whether the ABX or AX paradigm was adopted, due to poor performances in both the identification and discrimination tasks. Therefore, T2 and T4, which share the same contour shape, are actually perceived in a continuous manner.

It should be noted that although the heterotonic tonal continuum T2-T3 did not achieve the category perception mode in the strict sense, our results suggest that the perception of the heterotonic tonal continuum T2-T3 is more categorical than that of the homomorphic tonal continuum T2-T4. Additionally, our results revealed that there is no significant difference between ABX and AX paradigms.

5. References

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